

COURSE CODE	COURSE TITLE	L	T	P	E	C
UCS2603	THEORY OF COMPUTATION	3	0	0	0	3

OBJECTIVES

- To learn the mathematical notations and construct finite automata of the given language
- To understand and infer the relationship of finite automata and regular expressions
- To understand the language hierarchy, relationship between context free grammar and its language and to construct a PDA of CFG.
- To understand Turing machines and their programming capability
- To infer the knowledge of undecidable problems.

UNIT I MATHEMATICAL INTRODUCTION AND FINITE AUTOMATA 8

Basic Mathematical Notation and Techniques: Introduction to Formal Proof – Additional Forms of Proof – Inductive Proofs; Finite Automata (FA): Deterministic Finite Automata (DFA) – Non-deterministic Finite Automata (NFA) – Finite automata with epsilon transitions – Equivalence of FAs.

UNIT II REGULAR LANGUAGES 9

Regular Expressions and Languages: Regular expressions – Finite automata and regular expressions; Properties of Regular Languages: Proving languages not to be regular – Closure properties of regular languages – Equivalence and Minimization of Automata.

UNIT III CONTEXT FREE LANGUAGES AND PUSHDOWN AUTOMATA 10

Chomsky's Hierarchy of Languages; Context-Free Grammar and Languages: Context-Free Grammar (CFG) – Parse trees – Ambiguity in grammars and languages; Normal Forms for Context Free Grammars: Elimination of Useless Symbols, epsilon and unit productions – Chomsky Normal Form (CNF) – Greibach Normal Form (GNF). Pushdown Automata (PDA): Definition of the Pushdown automaton – The languages of a PDA – Equivalence of PDAs and CFGs – Deterministic Pushdown automata; Pumping Lemma for Context Free Languages.

UNIT IV TURING MACHINES 9

The Turing Machine – Programming Techniques for Turing Machines: Storage in the State – Multiple Tracks – Subroutines; Extensions to the Basic Turing Machine.

UNIT V UNDECIDABILITY 9

Undecidability: Language that is not Recursively Enumerable (RE) – Undecidable problem that is RE – Undecidable problems about Turing machines – Post's Correspondence Problem (PCP).

TOTAL PERIODS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Construct mathematical proofs related to computation and finite automata for a given language (K3)
2. Identify relationship between finite automata and regular expressions (K3)
3. Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)

4. Construct Turing machine for a given language (K3)
5. Explain the decidability or undecidability of various problems and make use of it (K3)

TEXTBOOKS

1. Hopcroft J E, Motwani R, Ullman J D, “Introduction to Automata Theory, Languages and Computations”, Pearson Education, 3rd Edition, 2008.

REFERENCES

1. Harry R Lewis, Christos H Papadimitriou, “Elements of the Theory of Computation”, Prentice Hall of India, 2nd Edition, 2003.
2. Peter Linz, “An Introduction to Formal Language and Automata”, Narosa Publishers, 3rd Edition, 2002.
3. Mishra K L P, Chandrasekaran N, “Theory of Computer Science – Automata, Languages and Computation”, Prentice Hall of India, 3rd Edition, 2004.

CO to PO Mapping

	PO 1	PO2	PO 3	PO 4	PO5	PO 6	PO7	PO 8	PO 9	PO10	PO1 1	PO12	PSO 1	PSO2
CO1	3	2											3	
CO2	3	2											3	
CO3	3	2											3	
CO4	3	2											3	
CO5	3	2								2			2	
Total	15	10								2			14	
Score	3	2								2			3	